ADRIEN ALLORANT

Lecturer in Social Statistics and Global Health

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PROFILE

I hold a joint appointment in the Department of Social Statistics and Demography and the Southampton Statistical Sciences Research Institute (S3RI), where I serve as Deputy Director. My research develops Bayesian, spatial, and survey methods for measuring health and inequality where data are sparse, contested, or expensive to collect. Three strands run through it: calibrating the routine data that countries already collect and own against small probability surveys; the production of official statistics under machine learning; and survey methodology and the measurement of inequality. Most of my applied work concerns HIV and immunisation coverage across African countries, in collaboration with the World Health Organization (WHO), UNAIDS, Gavi, UNICEF, and ministries of health.

ACADEMIC APPOINTMENTS

Lecturer in Social Statistics and Global Health, University of Southampton 2024–present

- Deputy Director, Southampton Statistical Sciences Research Institute (S3RI).
- Undergraduate Studies Lead, Department of Social Statistics and Demography (2026–present), responsible for the undergraduate module offer and programme coordination.
- Fellow of Advance HE (FHEA), awarded 2026 through the Southampton PREP route.

Visiting Assistant Professor in Statistics, Reed College, Portland, USA 2023–2024

- Taught introductory statistics and data science; supervised four undergraduate research and senior thesis students.
- Member of the Inclusive Pedagogy group, Center for Teaching and Learning.

Postdoctoral Research Fellow, McGill University and Imperial College London 2022–2023

- Led the UNAIDS Granular Data Analysis for Eastern and Southern Africa: secured USD 50,750, supervised three PhD students, and delivered 18 country reports informing Global Fund and PEPFAR applications.
- Advanced core components of Shiny90, the UNAIDS-sponsored model used by more than 40 countries for official HIV estimates, through analysis of programmatic and survey data across 41 countries.
- Facilitated UNAIDS regional training workshops in Abidjan for 17 Francophone ministries of health.

PhD Researcher and Graduate Research Associate, Institute for Health Metrics and Evaluation, University of Washington 2017–2021

- Led the geospatial analysis of tuberculosis surveillance in Bangladesh with the USAID Alliance for Combating TB; findings informed programme strengthening.
- Developed small-area estimation methods for health facility surveys, published in Nature Communications and now used at the Institute to assess disparities in healthcare quality.
- Contributed to the Institute's COVID-19 response team and co-authored four papers in The Lancet on global COVID-19 mortality, infections, and preparedness.

CONSULTANCY AND TECHNICAL ADVISORY

Technical Consultant, HIV Testing Team, World Health Organization, Geneva 2023, 2025–present

- Led the multi-country analysis of HIV self-testing scale-up across eight African countries.
- Developing the evidence base informing WHO guidance on HIV testing service delivery.

Technical Consultant, Health Inequality Monitoring, World Health Organization, Geneva 2023–2024

- Led the Gavi-commissioned landscaping review of spatial analysis methods for childhood immunisation coverage, co-authored with Gavi, the UNICEF geospatial lead, and WHO Immunisation.

PhD Research Associate, I-TECH and Haiti Ministry of Health 2018–2021

- Developed statistical methods for Haiti's national HIV electronic medical record (iSanté), now implemented as an automated performance-monitoring layer in the national health information dashboard.

Technical Consultant, Solthis, Paris and Guinea 2018–2019

- Led the impact evaluation of an HIV viral load testing intervention; built the R package ImpactSimul, used by the organisation for advocacy and grant applications.

PhD Summer Fellow, Institute for Disease Modeling, Gates Foundation 2020

EDUCATION

2017–2021 **PhD, Global Health Metrics**, University of Washington, Seattle

2015–2016 **MPH, Epidemiology**, Paris Saclay University (with distinction)

2013–2016 **Diplôme d'Ingénieur Statisticien-Économètre**, ENSAE ParisTech (with distinction)

2010–2013 **BSc Applied Mathematics; BA Social Sciences**, Université Paris I Panthéon-Sorbonne and Sciences Po Paris

RESEARCH FUNDING AND AWARDS

2026–2027 **Methodological review of centralised sampling and weighting**, Scottish Government. Co-Principal Investigator. **£40,000**

2026–2027 **Global Partnership Award (Africa)**, University of Southampton. Principal Investigator. **£9,016**

2025–2026 **Agreement for Performance of Work, HIV self-testing analysis**, World Health Organization. **£22,000**

2025 **Agreement for Performance of Work, Mozambique HIV self-testing monitoring**, World Health Organization. **USD 25,000**

2025–2026 **Small Research Grant, HIV programme pilot evaluation, Mozambique**, British Academy and Leverhulme Trust. **£10,000**

2025 **Web Science Institute internal funding**, University of Southampton. Ethnographic research on machine learning in quality frameworks for official statistics

2024–2025 **Research Culture Development Fund**, Faculty of Social Sciences, University of Southampton. Co-recipient with Professor Ann Berrington. **£1,015**

2023 **Granular Data Analysis of HIV Burden**, UNAIDS Eastern and Southern Africa. Principal Investigator. **USD 50,750**

2022–2023 **Postdoctoral Fellowship for Outstanding Foreign Students**, Fonds de Recherche du Québec Santé. **CAD 90,000**

2021 **Applied Population Health Research Fellowship**, University of Washington Population Health Initiative and King County Public Health. **USD 15,000**

2020 **PhD Summer Fellowship**, Institute for Disease Modeling, Gates Foundation. **USD 25,000**

2018–2019 **Impact evaluation model development**, French Agency for Development, via Solthis. **USD 25,000**

2017–2021 **Full PhD Scholarship**, Institute for Health Metrics and Evaluation. **USD 205,000**

Also awarded a CIHR Postdoctoral Fellowship (CAD 140,000), declined in favour of the Fonds de Recherche du Québec Santé award.

PEER-REVIEWED JOURNAL ARTICLES

- **Allorant A**, Bergen N, Danovaro-Holliday MC, Lorin J, Caetano Corrêa G, Boyda D, Belanger JL, Santhana Gopala Krishnan RS, Panciera R, Hosseinpoor AR. Assessing geographic inequalities in childhood immunisation coverage: a critical scoping review of spatial analysis methods. Vaccines. 14(7):572, 2026. doi:10.3390/vaccines14070572
- **Allorant A**, Smith PA. Algorithm-assisted inference and the future of official statistics. Journal of Official Statistics. 2026. doi:10.1177/0282423X261443590
- **Allorant A**, Bekelynck A, Monroe-Wise A, Baptista da Silva C, Chidarikire T, Edun O, Joaquim L, Kisia C, Larmarange J, Lyimo JJ, James JM, Musanhu C, Ncube G, Sathane I, Simo-Fotso A, Taasi G, Johnson CC. Scaling up HIV self-testing in Africa: insights from national programmatic data in eight countries. Journal of the International AIDS Society. 29(5):e70120, 2026. doi:10.1002/jia2.70120
- **Allorant A**. Adrien Allorant's contribution to the Discussion of 'Some statistical aspects of the COVID-19 response' by Wood et al. Journal of the Royal Statistical Society Series A: Statistics in Society. 189(1):59–60, 2026

- **Allorant A**, Kuchukhidze S, Stannah J, Xia Y, Masuku SS, Ekanmian GK, Imai-Eaton JW, Maheu-Giroux M. Socio-demographic and geographic disparities in HIV prevalence, HIV testing and treatment coverage: an analysis of 108 national household surveys in 33 African countries. Journal of the International AIDS Society. 28(8):e70024, 2025. doi:10.1002/jia2.70024
- Lyons C, Nambiar D, Johns NE, **Allorant A**, Bergen N, Hosseinpoor AR. Inequality in childhood immunization coverage: a scoping review of data sources, analyses, and reporting methods. Vaccines. 12(8):850, 2024
- Johns NE, Blumenberg C, Kirkby K, **Allorant A**, Dos Santos Costa F, Danovaro-Holliday MC, Lyons C, Yusuf N, Barros AJD, Hosseinpoor AR. Comparison of wealth-related inequality in tetanus vaccination coverage before and during pregnancy: a cross-sectional analysis of 72 low- and middle-income countries. Vaccines. 12(4):431, 2024
- **Allorant A**, Muset P, Hodgins C, Kirakoya-Samadoulougou F, Namachapa K, Mbofana F, Panagiotoglou D, Johnson LF, Imai-Eaton JW, Maheu-Giroux M. Temporal trends and determinants of HIV testing at antenatal care in Sub-Saharan Africa: a pooled analysis of population-based surveys (2005–2021). JAIDS Journal of Acquired Immune Deficiency Syndromes. 95(1S):e97–e105, 2024
- **Allorant A**, Fullman N, Leslie HH, Sarr M, Gueye D, Eliakimu E, Wakefield J, Dieleman JL, Pigott D, Puttkammer N. A small area model to assess temporal trends and sub-national disparities in healthcare quality. Nature Communications. 14:4555, 2023. doi:10.1038/s41467-023-40234-9
- Barber RM, Sorensen RJD, Pigott DM, Bisignano C, Carter A, Amlag JO, Collins JK, Abbafati C, Adolph C, **Allorant A**, et al. Estimating global, regional, and national daily and cumulative infections with SARS-CoV-2 through Nov 14, 2021: a statistical analysis. The Lancet. 399(10344):2351–2380, 2022
- Sorensen RJD, Barber RM, Pigott DM, Carter A, Spencer CN, Ostroff SM, Reiner RC, Abbafati C, Adolph C, **Allorant A**, et al. Variation in the COVID-19 infection-fatality ratio by age, time, and geography during the pre-vaccine era: a systematic analysis. The Lancet. 399(10334):1469–1488, 2022
- Bollyky TJ, Hulland EN, Barber RM, Collins JK, Kiernan S, Moses M, Pigott DM, Reiner RC Jr, Sorensen RJD, Abbafati C, **Allorant A**, et al. Pandemic preparedness and COVID-19: an exploratory analysis of infection and fatality rates, and contextual factors associated with preparedness in 177 countries. The Lancet. 399(10334):1489–1512, 2022
- Wang H, Paulson KR, Pease SA, Watson S, Comfort H, Zheng P, Aravkin AY, Bisignano C, Barber RM, Alam T, **Allorant A**, et al. Estimating excess mortality due to the COVID-19 pandemic: a systematic analysis of COVID-19-related mortality, 2020–21. The Lancet. 399(10334):1513–1536, 2022
- **Allorant A**, Biswas S, Ahmed S, Wiens KE, LeGrand KE, Janko MM, Henry NJ, Dangel WJ, Watson A, Blacker BF. Finding gaps in routine tuberculosis surveillance activities in Bangladesh. The International Journal of Tuberculosis and Lung Disease. 26(4):356, 2022
- Celestin K, **Allorant A**, Virgin M, Marinho E, Francois K, Honoré JG, White C, Valles JS, Perrin G, De Kerorguen N. Short-term effects of the COVID-19 pandemic on HIV care utilization, service delivery, and continuity of HIV antiretroviral treatment in Haiti. AIDS and Behavior. 25(5):1366–1372, 2021
- **Allorant A**, Parrish C, Desforges G, Robin E, Honoré JG, Puttkammer N. Closing the gap in implementation of HIV clinical guidelines in a low resource setting using electronic medical records. BMC Health Services Research. 20(1):804, 2020
- Prud'Homme C, Deschamps F, **Allorant A**, Massard C, Hollebecque A, Yevich S, Ngo-Camus M, Gravel G, Nicotra C, Michiels S. Image-guided tumour biopsies in a prospective molecular triage study (MOSCATO-01): what are the real risks?. European Journal of Cancer. 103:108–119, 2018
- Caramella C, **Allorant A**, Orlhac F, Bidault F, Asselain B, Ammari S, Jaranowski P, Moussier A, Balleyguier C, Lassau N. Can we trust the calculation of texture indices of CT images? A phantom study. Medical Physics. 45(4):1529–1536, 2018
- Naudet J, **Allorant A**, Ferry M. Heirs, corporate aristocrats and 'meritocrats': the social space of top CEOs and Chairmen in India. Socio-Economic Review. 16(2):307–339, 2018
- Naudet J, **Allorant A**, Ferry M. Héritiers, aristocrates d'entreprise et 'méritocrates': l'espace social du grand patronat en Inde. Entreprises et Histoire. 90(1):88–105, 2018
- Verlingue L, Malka D, **Allorant A**, Massard C, Ferté C, Lacroix L, Rouleau E, Auger N, Ngo M, Nicotra C. Precision medicine for patients with advanced biliary tract cancers: an effective strategy within the prospective MOSCATO-01 trial. European Journal of Cancer. 87:122–130, 2017
- Harttrampf AC, Lacroix L, Deloger M, Deschamps F, Puget S, Auger N, Vielh P, Varlet P, Balogh Z, Abbou S. Molecular screening for cancer treatment optimization (MOSCATO-01) in pediatric patients: a single-institutional prospective molecular stratification trial. Clinical Cancer Research. 23(20):6101–6112, 2017

MANUSCRIPTS UNDER REVIEW AND IN PREPARATION

- **Allorant A**, Amane G, Cumbana E, Magaia H, Mondlane J, Johnson C, Couto A, Gaspar I. HIV testing services in Mozambique one year after the PEPFAR stop-work order. BMJ Global Health. under review
- Youssif O, **Allorant A**. Reassessing the DHS wealth index as a proxy for socio-economic status: insights from population surveys in the Middle East and North Africa. PLOS ONE. under review
- **Allorant A**, Zhang L-C. Calibrating routine HIV testing data for subnational surveillance. Journal of the Royal Statistical Society Series A. in preparation

BOOK CHAPTERS, THESES, AND OTHER WRITING

- Naudet J, **Allorant A**, Ferry M. Heirs, corporate aristocrats, and 'meritocrats': the social space of top CEOs and chairmen in India. In: Mapping the Elite: Power, Privilege, and Inequality, Oxford University Press. p. 320, 2019
- **Allorant A**, Naudet J. The politics of antibiotics: an interview with Ramanan Laxminarayan. Books and Ideas. 2019
- **Allorant A**. Measuring inequities in health service delivery and their impact on health outcomes in low- to middle-income countries. PhD thesis, University of Washington. 2021

SELECTED INVITED TALKS

2026	Calibrating routine HIV testing data for subnational surveillance. Workshop, Italian National Institute of Statistics (ISTAT), Rome, Italy
2026	A statistical calibration framework for subnational health surveillance from routine data and survey audits. S3RI Research Festival, University of Southampton
2026	Statistical calibration of routine facility testing using probability-survey biomarkers to estimate undiagnosed prevalence. WorldPop weekly seminar, University of Southampton
2026	Statistical calibration of routine facility testing using probability-survey biomarkers to estimate undiagnosed prevalence. Spatio-Temporal Analyses Bayes seminar, Department of Biostatistics, University of Washington, Seattle, USA
2025	From routine data to sample placement: model-based adaptive design and joint estimation of HIV indicators in Mozambique. S3RI Research Festival, University of Southampton
2022	Temporal trends and major determinants of antenatal care HIV testing. Infectious Diseases Epidemiology Seminar, Imperial College London
2017	Heirs, corporate aristocrats and 'meritocrats': the social space of top CEOs and Chairmen in India. Socio-Economic Review conference on Elites, Economy and Society, Berlin, Germany

SELECTED CONFERENCE PRESENTATIONS

2026	Why ethnography? Studying AI adoption in official statistics as a problem of scientific practice. Royal Statistical Society International Conference, Bournemouth, United Kingdom
2026	Calibrating routine facility testing against probability-survey biomarkers for district-level estimation of undiagnosed HIV prevalence. Conference on Small Area Estimation (SAE 2026)
2025	Mismatched measures: comparing asset-based wealth index and income/consumption poverty in Mozambique. Italian Conference on Survey Methodology (ITACOSM), Bologna, Italy
2025	Re-assessing the DHS wealth index as a proxy for socio-economic status. Italian Conference on Survey Methodology (ITACOSM), Bologna, Italy
2025	Re-assessing the DHS wealth index as a proxy for socio-economic status in low resource settings. Population Association of America, Washington DC, USA
2023	Socio-demographic and geographic disparities in HIV burden and care engagement across African countries. ICASA, Harare, Zimbabwe
2022	A methodological framework to assess disparities in healthcare quality. Health Systems Global, Bogotá, Colombia
2018	Impact evaluation of HIV viral load testing intervention in Guinea. AIDS 2018, Amsterdam, Netherlands

WORKSHOPS AND CAPACITY BUILDING

2025	Routine health information systems for HIV surveillance. Week-long workshop for the Mozambique Ministry of Health monitoring and evaluation team, Maputo, funded by WHO
2023	UNAIDS regional training workshop on HIV estimates and projections. Facilitator for 17 Francophone ministries of health, Abidjan, Côte d'Ivoire

TEACHING

Undergraduate Studies Lead, Department of Social Statistics and Demography 2026–present

- Lead the department's undergraduate provision across two halves: the module offer, working with partner departments so that research methods and demography modules reach their students; and programme coordination, covering quality and reporting for the undergraduate degrees.
- Current work includes curriculum development for a revived population module and assessment redesign for a cohort with access to large language models.

RESM2001: Introduction to Social Data Analytics, University of Southampton, module lead 2025, 2026

- Year-2 foundation module of roughly 200 students, covering data wrangling, visualisation, and introductory modelling.
- Led the complete migration of the module from SPSS to R in 2025–26, including coursework, lectures, tutorials, assessment, and solutions, with a mid-semester curriculum redesign and new interactive tutorials built with learnr.

RESM3001: Applied Social Quantitative Analyses, University of Southampton, module lead 2024, 2025, 2026

- Final-year module built end to end in R and Quarto, covering measurement theory, sampling and survey design, regression, directed acyclic graphs and causal inference, models for categorical outcomes, and multiple imputation.

2023–2024	**MATH 241: Data Science** and **MATH 141: Introduction to Statistics and Probability**, Reed College, instructor
2022	**EPIB 638: Mathematical Modeling of Infectious Diseases**, McGill University, guest lecturer
2017–2021	**HMS 520**, **GH 345** and **GH 556**, University of Washington, teaching fellow and teaching assistant
2014–2016	**Micro-economic Evaluation of Public Policy**; **Introduction to Statistics and Econometrics**, Sciences Po Paris and Sorbonne, teaching assistant

SUPERVISION

2026–	**Clare Mitchell**, PhD, University of Southampton. Primary supervisor (60%), with Kristine Nilsen and Joseph Larmarange. The politics of indicators in global health, with a review of PEPFAR data and policies over the last ten years
2023–	**Dagim Fisseha Habteyesus**, PhD, University of Southampton. Co-supervisor (20%), with Kristine Nilsen and Sarah Neal. Sexual and reproductive health in refugee settings, Ethiopia
2022–2024	**Six undergraduate and master's students** at Reed College and McGill University, across summer research, senior theses, and global health scholarships

SERVICE AND PROFESSIONAL ACTIVITIES

2022–	**UNAIDS Reference Group on Estimates, Modelling and Projections**, member. Methods and models used by more than 40 African countries to produce official HIV estimates
2025–	**Royal Statistical Society**, International Development Section committee
2026	**Economic and Social Research Council**, assessor for the Advanced Quantitative Methods doctoral studentship track
2025–	**PhD Population Studies Masterclass**, co-organised with the London School of Economics and the ESRC South Coast Doctoral Training Partnership; leading the 2026 edition
2025	**Departmental away day on stakeholder and community engagement**, co-led with Professor Ann Berrington
2024–	**Peer review** for journals in statistics, global health, and epidemiology, five to ten reviews a year; departmental ethics review service

Fellow of the Royal Statistical Society. Member of the International Association of Survey Statisticians. Fellow of Advance HE.